

Show all of your work in the space provided to you. Full credit will not be awarded for correct answers unless sufficient supporting work is presented.

(16 pts) Let $A = \begin{bmatrix} 1 & -2 & -4 & 0 & -2 \\ 0 & 2 & 6 & 1 & 6 \\ 1 & 0 & 2 & 1 & 4 \\ 3 & 1 & 9 & 0 & 8 \end{bmatrix}$ and $\text{RREF}(A) = \begin{bmatrix} 1 & 0 & 2 & 0 & 2 \\ 0 & 1 & 3 & 0 & 2 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

a) What are the rank and the nullity of A

rank = 3

nullity = 2

b) Find a basis for the column space of A.

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -2 \\ 2 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

c) Find a basis for the null space of A.

$$x_1 = -2x_3 - 2x_5$$

$$x_2 = -3x_3 - 2x_5$$

$$x_3 = 1x_3 + 0x_5$$

$$x_4 = 0x_3 - 2x_5$$

$$x_5 = 0x_3 + 1x_5$$

Basis for null space is

$$\left\{ \begin{bmatrix} -2 \\ -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ -2 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$$

d) Is $\text{Col } A$ equal to $\mathbb{R}^4$? Give a reason for your answer.

No $\text{Col } A$ is a 3 dimensional subspace of $\mathbb{R}^4$.

2. (10 pts) Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ with $\det A = -2$. Find the following determinants.

a. $\det A^{-1} = -\frac{1}{2}$

b. $\det A^T = -2$

c. $\det A^3 = (-2)^3 = -8$

d. $\det(3A) = \overset{(3)}{3} \cdot (-2) = -54$ *Because A is 3x3 matrix*

e. $\det \begin{bmatrix} a & 5b & c \\ d & 5e & f \\ g & 5h & i \end{bmatrix} = 5 \cdot (-2) = -10$

3. (8 pts) Let A be any $n \times n$ invertible matrix. Use properties of determinants and inverses to prove that $A^T A$ is also invertible.

A is invertible so $\det(A) \neq 0$

$$\det(A^T) = \det(A)$$

$$\det(A^T \cdot A) = \det(A^T) \cdot \det(A) = [\det(A)]^2 \neq 0$$

Therefore $A^T \cdot A$ is invertible.

4. (6 pts) Let B be any 7×3 matrix with at least one nonzero entry.

a) What is the largest possible dimension of $\text{Col } B$?

3 since there are at most 3 pivots

b) What is the largest possible dimension of $\text{Nul } B$?

There are 3 columns and at least one nonzero column so at least one pivot column. There are at most 2 free variables so largest dim. for $\text{Nul } B$ is 2.

5. (10 pts) Let $B = \{V_1, V_2, V_3, V_4\}$ be a set of vectors in $\mathbb{R}^4$.

(4 pts) What two properties must B satisfy to be a basis for $\mathbb{R}^4$?

The set of vectors must be linearly independent and they must span all of $\mathbb{R}^4$.

(3 pts) Explain carefully why any set of fewer than 4 vectors cannot be a basis for $\mathbb{R}^4$

Fewer than 4 vectors cannot span all of $\mathbb{R}^4$

(3 pts) Explain carefully why any set of more than 4 vectors cannot be a basis for $\mathbb{R}^4$

More than 4 vectors in $\mathbb{R}^4$ will not be linearly independent.

6. (10 pts) Let $\mathbf{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$ with $\mathbf{b}_1 = \begin{bmatrix} -3 \\ 2 \\ -4 \end{bmatrix}$ and $\mathbf{b}_2 = \begin{bmatrix} 7 \\ -3 \\ 5 \end{bmatrix}$ be a basis for a subspace H in

$\mathbf{R}^3$ and let $\mathbf{x} = \begin{bmatrix} 5 \\ 0 \\ -2 \end{bmatrix}$. Find the coordinate vector for $\mathbf{x}$ relative to the basis $\mathbf{B}$.

$$\begin{bmatrix} -3 & 7 & : & 5 \\ 2 & -3 & : & 0 \\ -4 & 5 & : & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & : & 3 \\ 0 & 1 & : & 2 \\ 0 & 0 & : & 0 \end{bmatrix}$$

$$[\mathbf{x}]_{\mathbf{B}} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad \text{since} \quad \begin{bmatrix} 5 \\ 0 \\ -2 \end{bmatrix} = 3 \begin{bmatrix} -3 \\ 2 \\ -4 \end{bmatrix} + 2 \begin{bmatrix} 7 \\ -3 \\ 5 \end{bmatrix}$$

7. (10 pts) Let $A = \begin{bmatrix} 1 & -1 & 2 \\ 1 & -3 & 1 \\ 3 & 7 & 5 \end{bmatrix}$. Find an LU factorization for A and explain how the factorization is used to solve the equation $A\mathbf{x} = \mathbf{b}$.

$$\begin{bmatrix} 1 & -1 & 2 \\ 1 & -3 & 1 \\ 3 & 7 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 2 \\ 0 & -2 & -1 \\ 0 & 10 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 2 \\ 0 & -2 & -1 \\ 0 & 0 & -6 \end{bmatrix}$$

U

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 3 & -5 & 1 \end{bmatrix} \quad LU = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 3 & -5 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 \\ 0 & -2 & -1 \\ 0 & 0 & -6 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 2 \\ 1 & -3 & 1 \\ 3 & 7 & 5 \end{bmatrix}$$

To solve $A\vec{x} = \vec{b}$ write $L\underbrace{U\vec{x}}_{\vec{y}} = \vec{b}$

First solve $L\vec{y} = \vec{b}$ for $\vec{y}$

Then solve $U\vec{x} = \vec{y}$ for $\vec{x}$

8. (12 pts) Let T be any linear transformation from vector space V to vector space W . Recall that the kernel of $T = \{\mathbf{x} \text{ in } V : T(\mathbf{x}) = \mathbf{0}\}$. Prove that the kernel of T is a subspace of V .

Must show 3 things about $\ker T$

1. $\vec{0}$ is in $\ker T$ since $T(\vec{0}) = \vec{0}$

2. $\ker T$ is closed under vector addition.

Let $\vec{x}$ and $\vec{y}$ be in $\ker T$. [show $x+y$ is in $\ker T$.]

This means $T(\vec{x}) = \vec{0}$ and $T(\vec{y}) = \vec{0}$.

Therefore $x+y$ is in $\ker T$ because

$$T(x+y) = T(x) + T(y) = \vec{0} + \vec{0} = \vec{0}.$$

$\uparrow$
since T preserves addition

3. $\ker T$ is closed under scalar multiplication.

Let $\vec{x}$ be in $\ker T$ and c be any scalar.

[show $c\vec{x}$ is in $\ker T$]

$\vec{x}$ in $\ker T$ means $T(\vec{x}) = \vec{0}$.

$T(c\vec{x}) = cT(\vec{x})$ since T preserves scalar mult.

Therefore $c\vec{x}$ is in $\ker T$ because

$$T(c\vec{x}) = cT(\vec{x}) = c\vec{0} = \vec{0}.$$

9. (12 pts) Let T be the function from $\mathbf{P}^3$ to $\mathbf{P}^2$ that maps any polynomial, f , in $\mathbf{P}^3$ to its derivative f' in $\mathbf{P}^2$.

(9 pts) Prove that T is a linear transformation. You may use any facts from calculus that you know.

We must show that T preserves vector addition and scalar multiplication.

Let f and g be any polynomials in $\mathbf{P}^3$ and
Let c be any constant scalar.

$$\textcircled{1} T(f+g) = (f+g)' \underset{\substack{\uparrow \\ \text{By sum rule for derivatives}}}{=} f' + g' = T(f) + T(g)$$

$$\textcircled{2} T(cf) = (cf)' \underset{\substack{\uparrow \\ \text{Constant multiple rule for derivatives}}}{=} cf' = cT(f)$$

(3 pts) Describe all the functions in $\mathbf{P}^3$ that are in the kernel of T .

$T(f) = f'$. If $f' = 0$ then f must be
a constant function

$$f(x) = c \quad f'(x) = 0$$

10. (6 pts) Let A be an $n \times n$ matrix. Circle each letter beside any statement listed below that is equivalent to the statement " A is an invertible matrix."

☒ (a) There is a pivot in every row of $\text{RREF}(A)$.

☒ (b) There is a pivot in every column of $\text{RREF}(A)$.

☒ (c) The matrix equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for every vector $\mathbf{b}$ in $\mathbf{R}^n$.

☒ (d) A is row equivalent to the $n \times n$ identity matrix.

☒ (e) The span of the columns of A is equal to $\mathbf{R}^n$.

☒ (f) The determinant of A is not zero.

All of these are equivalent to
 A having an inverse.